

SANGRAM LEMBE

Pune, India

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🌐 [Linkedin](#)

🐙 [Github](#)

📁 [Portfolio](#)

EDUCATION

S.R.M. Institute of Science and Technology, Chennai

2025 - 2027

M.Tech - Data Engineering - CGPA - 8.39

Chennai, Tamilnadu

G.H.Raisoni College of Engineering and Management, Pune

2020 - 2024

B.Tech - Artificial Intelligence - CGPA - 8.31

Pune, Maharashtra

EXPERIENCE

Intern at CSIR-CSIO

March 2026 - Present

Role - Computer Vision Research Intern

Chennai, Tamilnadu

- Built **DroneSegNet**, a CNN-Transformer (PVT-v2) semantic segmentation model for UAV aerial imagery, achieving **76.18% mIoU** on the UAVid benchmark (8 classes) — a new **state of the art**, surpassing the prior best (75.49%).
- Designed a multi-term loss function (OHEM cross-entropy + Dice + boundary + self-supervised cross-scale consistency) and 12-pass test-time augmentation (TTA), improving small-object accuracy on rare classes.
- Ran single-factor ablations (attention family, depth, pretraining domain, rare-class augmentation): domain/altitude match — not model scale — drives accuracy; shipped a variant within 0.20 mIoU at **2.2× lower compute**.

Intern at Flourisense

July 2023 - January 2024

Role - Python Developer

Pune, Maharashtra

- Developed an Application Tracking System (ATS) using Node.js and Sequelize ORM to manage candidate data.
- Implemented and tested scalable RESTful APIs using Postman for seamless front-end and back-end integration.

PROJECTS

Nyaya-RAG — Era-Aware, Citation-Verified Legal RAG (Indian Statutes) [🔗](#) Jun 2026 - Present

- Built a fully local Retrieval-Augmented Generation (RAG) system over **17 Indian statutes** (vLLM Qwen2.5-14B, BGE-M3, Qdrant vector database — zero external APIs), modeling India's 2024 IPC → BNS criminal-law transition via era-tagged chunks and cross-era section mapping.
- Cut hallucinated legal citations from 31.7% to **0%** (citation precision 0.68 → 1.000) via a post-generation citation verifier validating every cited section against the corpus; achieved 1.000 era-correctness on criminal-law queries.
- Designed a 150-question grounded gold set and evaluation harness; retrieval ablations (dense vs BM25 vs hybrid RRF) achieved **recall@10 = 1.00** and **MRR = 0.92** with BGE-M3 embeddings.

LogiCrisis — Meta PyTorch OpenEnv Hackathon Finalist [🔗](#)

Apr 2026

- Built a six-agent Reinforcement Learning (RL) environment simulating cascading supply-chain crises across India using live weather, currency, and GDELT conflict APIs — rated "top 0.1% of RL developers" by a Meta engineer.
- Fine-tuned Llama-3-8B (GRPO + Unsloth 4-bit QLoRA on A100 GPU), yielding emergent multi-agent behaviors: pre-emptive cold-storage deployment, two-turn-early geopolitical alerts, and counter-proposals to currency shocks.
- Selected for the 48-hour Bangalore finale from **51K+ applicants** after qualifying with a Banking Support Agent environment for fraud dispute resolution.

Forecasting the Psychological Well-being of Students Using ML [🔗](#)

Dec 2025 - Mar 2026

- Awarded **Gold Medal** at College Research Day for Best Research Paper from the Department of Psychology.
- Developed a predictive machine learning model utilizing PCA, PSO, and stacked ensemble techniques to forecast mental health challenges among 1,000 college students.
- Enabled early identification of at-risk students, supporting targeted interventions to improve well-being.

Technical Skills

Languages: Python, C++, SQL, HTML, CSS

ML/DL: PyTorch, TensorFlow, Scikit-learn, CNNs, Vision Transformers (PVT-v2), Semantic Segmentation, Reinforcement Learning (GRPO), Fine-tuning (QLoRA, Unsloth), Mixed Precision Training

LLMs & GenAI: RAG, LangChain, FAISS, Qdrant, vLLM, HuggingFace, Multi-Agent Systems, BM25 / Hybrid Retrieval (RRF), Embeddings (BGE-M3), LLM Evaluation, Prompt Engineering

Tools & Visualization: Git, VS Code, PyCharm, Anaconda, Postman, n8n, Jupyter, Canva, Matplotlib, Plotly

PUBLICATIONS

- Features Preserving Blurred Image Classification Using Large Language Model** [🔗](#) — In IJIRSET, pp. 8474–8480, 5 May 2024, Pune. DOI: 10.15680/IJIRSET.2024.1305313

ACHIEVEMENTS & CERTIFICATIONS

- Awards:** Meta PyTorch OpenEnv Hackathon Finalist (2026); Gold Medal – Best Research Paper, SRM Research Day; First-author paper submitted to ACCV 2026

Certifications: Fundamentals of Deep Learning (NVIDIA); The Joy of Computing Using Python (NPTEL)